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Author for correspondence:

G.J. Cooke

e-mail: gjc53@cam.ac.uk

The development of the ozone layer under an evolving Sun

A. Kumar¹, G.J. Cooke¹

¹Institute of Astronomy, University of Cambridge,
Cambridge, CB3 0HA, UK

Earth's atmospheric evolution has been inferred from multiple lines of geological evidence. Its atmosphere transitioned from a composition devoid of molecular oxygen (O₂) during the Archean eon (4 – 2.4 Gyr ago) to one where O₂ was abundant and is now 21% of our modern atmosphere by volume. With high quantities of O₂, it becomes possible for an ozone (O₃) layer to form, partially shielding the surface from ultraviolet radiation which is harmful to life. The evolution of the O₃ layer and its corresponding thickness has previously been simulated under the modern Sun with boundary conditions similar to present day. Here, we thoroughly explore the impact of a younger Sun and various boundary conditions relevant to different geological periods, using four open-source one-dimensional (1D) photochemical models to predict the abundances of multiple atmospheric species, focusing on O₃. We run thousands of simulations of Earth's atmosphere under various oxygenation states relevant to its history and benchmark these results against results from a 3D chemistry climate model. We show that the ozone layer thickness is model dependent, and is sensitive to the evolving solar spectrum, pressure-temperature profiles, surface albedo, solar zenith angle, and atmospheric methane concentrations. *Many of these parameters are interlinked, conflating... In particular, the differences in 1D models are due to... Our work is relevant to exoplanets, where the data in comparison to Earth is more uncertain with fewer measurable parameters. We quantify the effects of model choices and choices of boundary conditions on observational predictions by generating spectra for Earth-like exoplanets.*

1. Introduction

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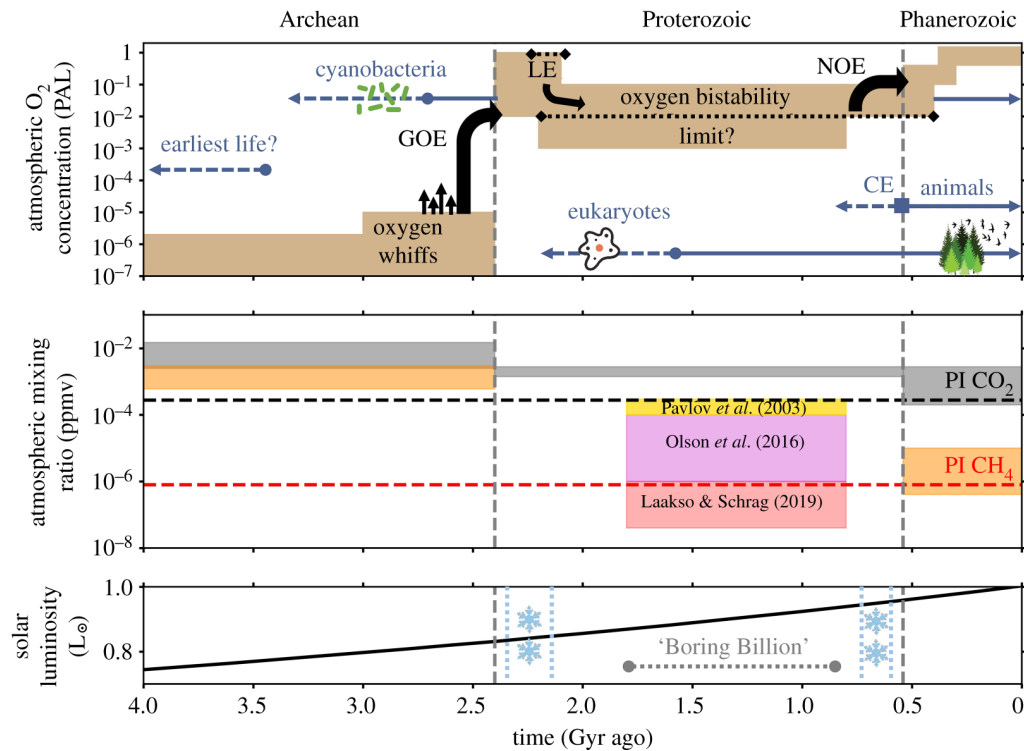


Figure 1. To be changed - here right now asd a placeholder

Over the course of Earth's history, its atmosphere has undergone a profound transformation, from an anoxic (O_2 -poor) state during the Archean eon (4 – 2.4 Ga, where Ga is billions of years ago), to the O_2 -rich (21% by volume) atmosphere animals inhabit today (Catling and Zahnle, 2020). A possible influencing factor in Earth's biological history is the formation and thickness of the ozone (O_3) layer which depends, in part, on the atmospheric O_2 concentration. O_3 molecules absorb biologically harmful ultraviolet (UV) radiation, thus shielding life on the surface. While terrestrial life enjoys the benefits of a modern ozone layer, it has not always been present throughout Earth's past.

Earth's surface and atmospheric oxygenation occurred in several key stages. The first of these was at the end of the Archean, during one of the most significant environmental shifts in our planet's history - the Great Oxidation Event (GOE; Lyons et al., 2014). This ~200 Myr transition started around 2.4 Ga (Poulton et al., 2021) and resulted in atmospheric O_2 levels rising from less than 10^{-6} of the Present Atmospheric Level (PAL) to between 1% and 10% PAL (Bekker et al., 2004; Holland, 2006). Evidence for the GOE arises from a myriad of sources (Catling and Zahnle, 2020), which include: the disappearance of mass-independent fractionation of sulfur isotopes (Farquhar et al., 2000); the cessation of atmospheric xenon fractionation (Zahnle et al., 2019); the appearance of continental red beds (Shawwa et al., 2024); the termination of widespread Banded Iron Formations (BIFs) (Bekker et al., 2010; Yin et al., 2023); the disappearance of easily oxidized minerals like detrital uraninite and pyrite in sediments (Johnson et al., 2014); and the presence of oxidised palaeosols post-GOE (Babechuk et al., 2019).

Following the GOE, atmospheric O₂ dropped again (Bekker and Holland, 2012; Canfield et al., 2013; Lyons et al., 2014) at the start of the Proterozoic eon (2.4 – 0.541 Ga). Much of the Proterozoic saw a prolonged period of apparent environmental stability, known as the “Boring Billion” (Buick et al., 1995). Atmospheric O₂ levels during this period remained relatively stable at around 10^{−3} to 10^{−1} PAL (Lyons et al., 2021, 2014). While a stasis may characterise the Mesoproterozoic in terms of atmospheric redox, carbon isotope excursions, supercontinent cycles (Westall and Xiao, 2024), and a constant day length (Mitchell and Kirscher, 2023), the era was hugely important in terms of biological evolution (Westall and Xiao, 2024).

The late Proterozoic likely underwent a second rise in atmospheric O₂, in an event dubbed the Neoproterozoic Oxidation Event (NOE; Och and Shields-Zhou, 2012), evidenced by shifts in carbon isotope values (Krissansen-Totton et al., 2015) and other redox proxies (Lu et al., 2017; Pogge von Strandmann et al., 2015; Turner and Bekker, 2016). The atmospheric O₂ level following the NOE is still debated, but estimates range from between 10 and 50% PAL (Blamey et al., 2016; Steadman et al., 2020; Yeung, 2017). In the subsequent Phanerozoic eon (0.541 Ga – present), O₂ levels are understood to have fluctuated between 10% and 150% PAL, first reaching approximately modern day concentrations during the Paleozoic (Lyons et al., 2021; Wallace et al., 2017).

Parallel to this dramatic evolution of our atmosphere, the Sun has also evolved significantly over Earth’s history. Models of stellar evolution predict that the young Sun was only 70–75% as bright as the modern Sun (Gough, 1981; Newman and Rood, 1977). The Faint Young Sun Paradox (Feulner, 2012) describes the discrepancy between the early Sun’s reduced luminosity and the geological evidence for liquid water on Earth’s surface, because ice-albedo feedback should result in increased reflection of incoming radiation and lower temperatures. This feedback loop could theoretically result in a runaway glaciation and global ice cover, leading to a “Snowball Earth” state, where the entire planet is covered in ice (Kirschvink and Schopf, 1992). However, this scenario is in conflict with evidence of liquid water, including 4.4 Ga old zircon grains (Mojzsis et al., 2001), 3.8 Ga old pillow basalts and sedimentary rocks (Furnes et al., 2004; Lowe, 1980), and evidence for microbial life from 2.5 – 3.5 Ga old microfossils and stromatolites (Noffke et al., 2013; Schopf, 2006; Schopf et al., 2007). While climatic models have largely reconciled this paradox for the Archean (Charnay et al., 2020; Goldblatt et al., 2021), the mechanisms sustaining ice-free conditions throughout the Proterozoic remain poorly understood (Cooke et al., 2022; Fiorella and Sheldon, 2017; Ji et al., 2023; Olson et al., 2016). *Earth’s geological, atmospheric, and biological history, including uncertainties, is summarised in Figure 1.*

Many of the aforementioned astrophysical, geological, and biological factors are likely to have influenced the thickness of the O₃ layer. The thickness is also referred to as the column depth (given in terms of molecules per area in a vertical column above the surface), which varies with season and position above the Earth’s surface due to atmospheric chemistry, temperature differences, and the Brewer-Dobson circulation (Butchart, 2014). A thorough understanding of the O₃ layer’s temporal variation and its influence on biological habitats has historically been derived from one-dimensional (1D) atmosphere modelling studies (Kasting and Donahue, 1980; Kasting et al., 1985). Only a few studies (Cooke et al., 2022; Jaziri et al., 2022; Ji et al., 2023) have compared these 1D models to 3D coupled chemistry-climate models. Once considered a ‘solved problem’ (Kasting, 2025), the co-evolution of the O₃ layer with atmospheric O₂ concentrations is now quantitatively unclear because no single model (1D or 3D) accounts for all the different influencing parameters together (Cooke et al., 2022; Ji et al., 2023; Ji et al., 2024; Kasting, 2025; Yassin Jaziri et al., 2022).

Investigating the discrepancies between various 1D and 3D models is crucial. 1D photochemical models are widely used to study the habitability of early Earth-like and Earth-like exoplanets (Meadows et al., 2023; Rugheimer and Kaltenegger, 2018; Rugheimer et al., 2015; Segura et al., 2003; Wunderlich et al., 2020). Why the disparity in predictions arise is muddled by

differences in assumptions, model setups, chemical reaction rates, vertical mixing, temperature-pressure profiles, and boundary conditions, which are not always thoroughly reported in various works. This makes it difficult to unravel where the fundamental disagreements emerge from.

In this work, we aim to disentangle these complex phenomena and characterize their broader implications for early-Earth and exoplanet modelling by using four open-source photochemical models (some of which are widely used by the exoplanet community). We run thousands of simulations Earth's atmosphere under a range of conditions pertinent to its geological history. Focussing on the development of the O₃ layer, we systematically vary the incoming stellar spectrum to account for an evolving Sun, and vary the atmospheric O₂ mixing ratio and the largely unknown pressure-temperature profiles of the post-GOE Earth. We explore model sensitivities to a range of additional input parameters to explore how various assumptions affect atmospheric profiles of key species. We compare results from these models to each other, and to outputs from a 3D chemistry-climate model. Finally, we comment on whether these models are “tuned” to reproduce the modern observed atmosphere of Earth, with this tuning propagating through as a fundamental error to paleoatmosphere calculations.

2. Atmospheric computational modelling

Numerical models of planetary atmospheres are useful tools for understanding the chemical composition, evolution, and climate of planetary atmospheres, ranging from Earth's modern anthropogenic climate change (Danabasoglu et al., 2020; Tokarska et al., 2020) to unravelling the mysteries of exotic exoplanets (Shields, 2019; Wogan et al., 2025a).

1D photochemical models construct the atmosphere as a single vertical column, predicting the variation of chemical species with altitude (or pressure). Usually, a stellar spectrum is input at a fixed solar zenith angle which is chosen to account for the diurnal cycle or to get the average climate response. These models can be coupled to a climate model, where the temperature profile is adjusted based on the chemical composition, albedo and incoming radiation. 1D photochemical models solve the continuity equation which calculates chemical production and loss with vertical transport. Vertical mixing is parameterised through eddy diffusion (K_{zz}) and molecular diffusion. Because 1D models are computationally efficient, they can incorporate highly complex chemical networks involving hundreds of species and thousands of reactions. They are useful for exploring fundamental chemical pathways, calculating global mean states, and conducting long-term evolutionary studies, such as the rise of oxygen on early Earth or the detection of trace gases in planetary atmospheres.

3D Atmospheric chemistry-climate models, expand on 1D models to include horizontal transport and spatial heterogeneity, where each atmospheric grid cell has a solar zenith angle that depends on the planetary system's orbital parameters. Unlike 1D models, 3D frameworks capture the influence of atmospheric circulation, such as winds and atmospheric waves (e.g. Kelvin waves and gravity waves), and the day-to-night (diurnal) cycle, which are essential for predicting spatial and temporal variations, including seasonal effects on Earth. Therefore, 3D models have the potential to provide a more realistic representation of an atmosphere. Unfortunately, they are significantly more computationally expensive and cannot always include as many chemical processes as 1D models in order to remain feasible for research and forecasting purposes.

In this section, we introduce the four open-source 1D photochemical models *Atmos*, *VULCAN*, *Photochem*, and the *Kasting 1D* model. We also introduce the open-source 3D chemistry-climate model *WACCM6*.

(a) *Atmos*

*Atmos*¹ is an open source one-dimensional coupled photochemistry and climate code (Arney et al., 2016; Virtual Planetary Laboratory, 2021). It can generate self-consistent atmospheric compositions and temperature-pressure profiles, under varying solar evolution, atmospheric

¹ *Atmos*: <https://github.com/VirtualPlanetaryLaboratory/atmos>

boundary conditions, and input parameters. The photochemical model in Atmos draws from a model developed by [Kasting et al. \(1979\)](#), updated by [Zahnle et al. \(2006\)](#), and described in detail by [Arney et al. \(2016\)](#) and ([Meadows et al., 2018](#)). It divides the atmosphere into plane-parallel layers in hydrostatic equilibrium, with a vertical transport scheme including molecular and eddy diffusion. Atmos solves the coupled continuity and flux equations using a fully implicit reverse Euler method, which reduces to Newton's method as $t \rightarrow \infty$. Implicit Jacobian iteration is used to maintain stability and convergence in stiff chemical systems. For species that are nearly decoupled or negligibly abundant (e.g., S_8 aerosol), Atmos uses a reduced tri-diagonal inversion scheme to solve transport independently from the main Jacobian. Boundary fluxes can be specified as either constant mixing ratios, fixed fluxes, or deposition velocities. The model uses the δ -2 stream method for radiative transfer [Toon et al. \(1989\)](#). We used the "Modern Earth Complex" template described in [Lincowski et al. \(2018\)](#), which is an updated version of the "Modern Earth" template from [Meadows et al. \(2018\)](#). The updated version includes a higher resolution wavelength grid with 309 reactions and 73 species, and has been validated against Earth and Venus in Appendix C of [Lincowski et al. \(2018\)](#). The climate portion of the model was developed by [Kasting and Pollack \(1983\)](#), [Kasting et al. \(1984b\)](#), [Kasting et al. \(1984a\)](#) and [Kasting and Ackerman \(1986\)](#), and was updated by [Kopparapu et al. \(2013\)](#) and [Arney et al. \(2016\)](#). It also uses the δ -2 stream approximation for radiative transfer, and uses the correlated-k coefficients to calculate the spectral absorption of O_2 , O_3 , CO_2 , H_2O , CH_4 , and C_2H_6 across 38 spectral intervals. Infrared fluxes are calculated with the RRTM routine ([Mlawer et al., 1997](#)), employing 16-term sums in each spectral band for calculated k-coefficients. The tropospheric temperature profile is derived by tracing a moist adiabatic lapse rate up to the altitude where the stratospheric temperature is attained ([Kasting, 1988](#)), unless the atmosphere is desiccated, in which case a dry adiabat is used instead. The profile from [Manabe and Wetherald \(1967\)](#) is used to determine the water vapour distribution profile, assuming a surface humidity of X% [Kasting and Ackerman \(1986\)](#). The two models can be coupled by running them back and forth, with the photochemical model providing species profiles to the climate model, and the climate model providing the pressure temperature profile. **What does it do for solar zenith angle, clouds, albedo, lightning, and flux options with altitude?**

(b) Photochem

Photochem² is an open source one-dimensional coupled climate–photochemistry model derived from atmos ([Wogan et al., 2025a, 2023](#)). It has been used to study the atmospheres of a wide range of bodies, including the Early Earth ([Wogan et al., 2023](#)), Gas Giants ([Mukherjee et al., 2025](#)), sub-Neptunes ([Benneke et al., 2024](#); [Schmidt et al., 2025](#); [Wogan et al., 2024](#)), brown dwarfs ([Beiler et al., 2024](#); [Wogan et al., 2025b](#)), and several solar system bodies, including Venus, Earth, Mars, Jupiter and Titan ([Wogan et al., 2025a](#)). Photochem contains a 1-D photochemical model, a 1-D climate model and an equilibrium chemistry solver, all detailed in [Wogan et al. \(2025a\)](#). The photochemical model solves the coupled continuity equations for molecular transport and chemical kinetics. An Eddy diffusion coefficient (K_{zz}) is used to parametrize vertical mixing. The reaction network includes 610 forward reactions (need to double check this) and 95 photolysis reactions, containing 98 gas-phase species composed of H, O, C, N, A, Cl and He. . The climate module solves for atmospheric convective and radiative energy balance. **What does it do for solar zenith angle, clouds, albedo, lightning, and flux options with altitude?**

Photochem uses high resolution O_2 cross sections from [Heays et al. \(2017\)](#), who source their high-resolution O_2 cross sections in the SRB from [Yoshino et al. \(1992\)](#). It includes absorbance in the SRB for CO_2 (from [Schmidt et al. \(2013\)](#)) and H_2O (from [Heays et al. \(2017\)](#), sourced from [Mota et al. \(2005\)](#)). It calculates Rayleigh scattering using the formalism of [Vardavas and Carver \(1984\)](#). The required data are compiled from several sources: [Keady and Kilcrease \(2000\)](#) and [Penndorf \(1957\)](#) for N_2 , O_2 , H_2 , He, and CO; [Batalha et al. \(2019\)](#) for CH_4 and NH_3 ; and [Ranjan et al. \(2017\)](#), [Keady and Kilcrease \(2000\)](#), and [Murphy \(1977\)](#) for H_2O .

²<https://github.com/Nicholaswogan/photochem>

(c) VULCAN

We used VULCAN³, an open source one-dimensional photochemical kinetics code for planetary atmospheres which has been validated has been benchmarked against several models including TEA (Blecic et al., 2016) and ARGO (Rimmer and Helling, 2016) in Tsai et al. (2017) and Tsai et al. (2021). VULCAN solves the Eulerian continuity equation for multiple species and accounts for production, loss and vertical transport. We implement VULCAN with the equilibrium chemistry code FastChem⁴, which calculates gas phase equilibria (Stock et al., 2022, 2018). We use the updated full S-N-C-H-O photochemical reaction network for oxidizing atmospheres⁵.

(d) Kasting 1D model

The Kasting-group 1D photochemical model, a widely used one-dimensional framework for simulating Earth and exoplanet atmospheres (Ji et al., 2023; Ji et al., 2024; Kasting and Donahue, 1980; Kasting et al., 1979; Segura et al., 2003) has been used and updated over 4 decades. The model computes vertical profiles of atmospheric composition by solving continuity equations with prescribed boundary conditions, incorporating photochemistry and vertical transport.

Photolysis of O₂ in the Schumann–Runge (SR) bands (175 – 205 nm) was originally parameterized following (Allen and Frederick, 1982), which divided the SR region into 17 wavelength intervals based on quantum mechanical calculations (Frederick and Hudson, 1980a,b) and laboratory measurements (Hudson and Carter, 1968). The original implementation used a band model valid for present-day O₂ levels, with absorption occurring primarily in the upper stratosphere and mesosphere. To treat multiple scattering in low-O₂ atmospheres, the model adopted the (Toon et al., 1989) δ -2-stream method and replaced the older (Hudson and Mahle, 1972) parameterization with the Allen and Frederick (1982) band model. Exponential sum fits were derived from Allen and Frederick (1982) to couple with the scattering treatment, though these lacked explicit temperature dependence and thus performed poorly under low-O₂ conditions.

In recent updates, the model's O₂ photolysis scheme has been revised to include a correlated-k parameterization based on line-by-line calculations (Fernández et al., 2007; Ji et al., 2024). This approach accounts for the strong temperature dependence of SR absorption, includes overlapping continua (the Schumann–Runge continuum and the Herzberg continuum), and performs reliably across a wide range of O₂ levels. Comparisons with laboratory data (Yoshino et al., 1992) and the WACCM6 parameterisation show that the correlated-k method improves accuracy for both modern and low-O₂ atmospheres when multiple scattering is included.

However, the new update to the Kasting 1D model (Ji et al., 2024; Kasting and Ji, 2025) overpredict the amount of O₃ for the modern atmosphere at 330 Dobson Units versus 292 DU (Braesicke et al., 2019). This implies one of three things, that either: the Kasting 1D model is missing some important processes (physics/chemistry) that would result in a more accurate O₃ column; that the model was originally tuned for the O₃ column and need re-tuning; or that 3D models are required to accurately predict O₃ concentrations. In addition, Photochem, Atmos, and VULCAN do not employ the up-to-date accurate parameterisation of O₂-photolysis in the Schumann–Runge bands. It is because of this that we suspect that all open-source 1D models that are widely used for paleo-atmospheric science and exoplanet science regarding oxygenated atmospheres, are likely in error. This forms the premise of our work.

(e) WACCM6

We use 3D chemistry-climate model simulations of the pre-industrial and early Earth presented in Cooke et al. (2022), where the Earth System Model WACCM6 was used to model different

³VULCAN: <https://github.com/shami-EEG/VULCAN>

⁴FastChem: <https://github.com/NewStrangeWorlds/FastChem>

⁵S-N-C-H-O Network: https://github.com/shami-EEG/VULCAN/blob/master/thermo/SNCHO_full_photo_network.txt

oxygenation levels relevant to Earth's geological history (Gettelman et al., 2019). WACCM6 is a configuration of the Coupled Earth System Model (CESM), and Cooke et al. (2022) used version CESM2.1.3⁶ with minor changes in the initial conditions⁷, which are described in Cooke et al. (2022) and summarised here.

The pre-industrial simulation is representative of the Earth's atmosphere circa 1850, along with modern day ocean and land settings. Cooke et al. (2022) set the spatial resolution at 1.875° by 2.5° (96 latitude and 144 longitude points), containing 70 vertical atmospheric layers from 1000 hPa to 4.5×10^{-6} hPa (approximately 150 km in altitude). Cooke et al. (2022) ran 8 simulations by varying the amount of atmospheric O_2 between 0.1 – 150% PAL, encompassing estimates for the Earth mentioned in section 1. The abundance of the background gas N_2 was changed to maintain atmospheric pressure. To account for uncertainty in estimates of both the amount of atmospheric O_2 and temporal uncertainties in when exactly O_2 rose in the Earth's atmosphere, simulations covered the entire range, including PI (Pre-Industrial), 150% PAL, 50% PAL, 10% PAL, 5% PAL, 1% PAL, 0.5% PAL, and 0.1% PAL. Surface boundary conditions were maintained while O_2 was altered. This, as noted by Cooke et al. (2022), is unrealistic, but isolates the effect of changes in oxygenation to atmospheric chemistry and dynamics.

WACCM6 includes water vapour microphysics (e.g. evaporation, rainout, clouds), but doesn't account for hydrogen escape and water loss, which can be calculated through post-processing once quasi-steady-state has been reached (Cooke et al., 2025). WACCM6 also includes atmosphere-ocean interactions, producing a more realistic measure of dynamic responses to changes in environmental conditions (like temperature). The chemical network includes 98 species and 298 reactions and is described in Emmons et al. (2020). The simulations in Cooke et al. (2022) used an atmospheric time step Δt of 30 minutes, calculating the concentration of 75 species implicitly, and 22 species explicitly, holding N_2 constant (Sandu et al., 1997). Simulations did not include tropospheric O_3 production due to volatile organic compounds. For each scenario, the simulation was run for 30 to 90 years until the trend in total hydrogen atoms stabilized in the middle atmosphere (12–100 km), with the outputs averaged over the last 4 simulated years.

(f) Radiative Transfer Parametrisations in the Schumann-Runge Bands

The accuracy of photochemical models depend entirely on how they calculate the penetration of solar radiation to determine photolysis rates. This is particularly challenging in the Schumann-Runge bands (SRB, 175–205nm), where O_2 absorption cross sections vary by orders of magnitude over very short wavelength intervals, and exhibit strong temperature dependence. Table 1 summarises the parametrisation schemes used by various models to calculate UV transmission in the SRB.

Table 1. Chlorine chemistry?

Model	Temperature dependence in SRB	Absorbance of CO_2/H_2O in SRB	Scattering in SRB
Atmos	?	?	?
Photochem	No	Yes	Yes
VULCAN	Probably	?	?
Kasting 1D model	Yes, correlated-k (Ji et al., 2024)	Yes	Yes, ✓, source
WACCM6	Yes, Koppers and Murtagh	Added in	No

⁶CESM2.1.3: <https://www.cesm.ucar.edu/models/cesm2>

⁷ExoCESM Github: https://github.com/exo-cesm/CESM2.1.3/tree/main/O2_Earth_analogues

3. Simulations performed

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In this section, we describe the modelling choices used to benchmark 1D model predictions against 3D chemistry-climate data. To ensure that divergences in model outputs are driven by intrinsic differences (such as differing chemical networks, transport schemes, and numerical solvers) rather than inconsistent inputs, we utilise uniform boundary conditions that mirror our WACCM6 pre-industrial control simulations (described in [Cooke et al. \(2022\)](#) and [Cooke et al. \(2023\)](#) where possible. At the lower-boundary, these were fixed mixing ratios⁸ and surface fluxes⁹ shown in Table 2. While WACCM6 simulations incorporated temporally and spatially varying CESM CMIP6 emissions from the year 1750 cycled over a 12 month period, our 1D runs incorporate static global averages to represent annual mean fluxes, with the total sum used for species having multiple sources (e.g., biomass burning, oceans, soils etc). For models that do not support altitude-dependent volcanic injection of SO₂, we apply the aggregate flux in Table 2 as a single lower-boundary condition.

Table 2. WACCM6 boundary conditions adopted for our 1D simulations. Values represent a hybrid of fixed boundary conditions and CESM2 emission fluxes.

Species	Mixing Ratio	Flux (molecules cm ⁻² s ⁻¹)
<i>Fixed Mixing Ratios (Cooke et al. (2022))</i>		
N ₂	0.78	-
O ₂	0.21	-
CH ₄	0.808×10^{-6}	-
CO ₂	284×10^{-6}	-
H ₂	5.00×10^{-7}	-
OCS	3.37×10^{-10}	-
N ₂ O	2.73×10^{-7}	-
<i>Emission Fluxes (CESM2 CMIP6)</i>		
CO	-	4.98×10^{10}
NO	-	2.37×10^9
CH ₂ O	-	5.16×10^8
SO ₂	-	1.43×10^9

To capture the broad parameter space required for our analysis, we varied three key inputs: the atmospheric oxygenation state, the pressure-temperature (P-T) profile, and stellar spectra. In all models, atmospheric oxygenation was varied between 150%, 100%, 10%, 1% and 0.1% PAL, to bracket the uncertainties surrounding the geologic history of atmospheric oxygen. We used corresponding P-T profiles generated self-consistently from [Cooke et al. \(2022\)](#) (figure 2).

⁸WACCM6 fixed boundary conditions : https://github.com/exo-cesm/CESM2.1.3/blob/main/O2_Earth_analogues/README.md

⁹CESM CMIP6 emissions : <https://github.com/lkemmons/cesm-cmip6-emissions>

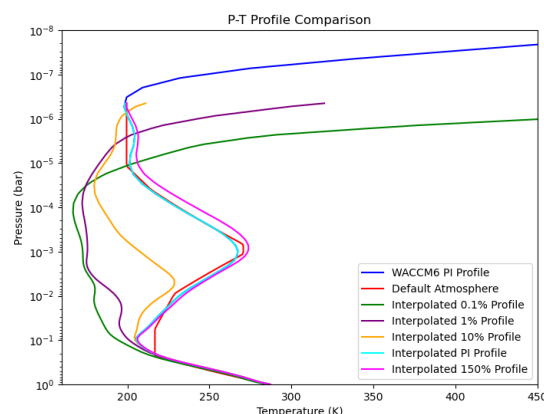


Figure 2. Self Consistent P-T profiles relevant to various oxygenation states of the Earth generated from WACCM6. plot needs to be changed

To model atmospheres under a young sun, we use parametrized solar fluxes from [Claire et al. \(2012\)](#) as model inputs. These fluxes represent the solar output of the modern sun and the sun at stellar ages of 0.5, 1, 1.5, 2 and 2.4 Ga. Figure 3 shows the total solar flux, and fluxes in bands relevant to O_3 photochemistry (i.e, the Schumann-Runge Continuum, the Schumann-Runge Bands and the Herzberg Continuum). These spectra were manually used as inputs in photochem, while Atmos has an inbuilt function to calculate altered spectra using methods in [Claire et al. \(2012\)](#) for varying solar ages.

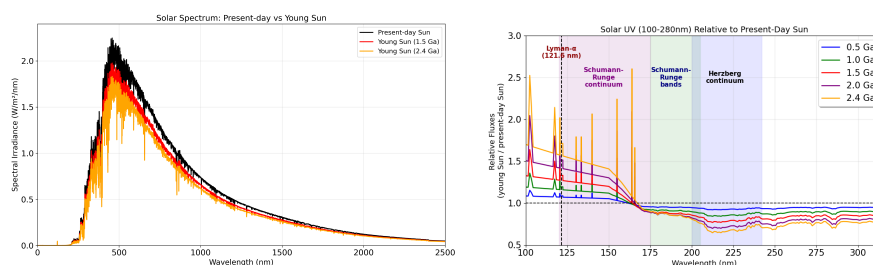


Figure 3. Top: Irradiance vs Wavelength for various solar ages. Bottom : Evolution of solar UV fluxes relative to the present-day Sun (0 Ga) for 0.5–2.4 Ga. Shaded regions mark the Schumann–Runge continuum (< 175 nm), Schumann–Runge bands (175–205 nm), and the Herzberg continuum (200–242 nm) that became shielded by O_3 after the Great Oxidation Event

Do we need to say something like this? [Wogan et al. \(2025a\)](#) describe Photochem an updated version of Atmos. Because it produces outputs consistent with VULCAN but offers lower runtimes, we utilized Photochem to explore the majority of the parameter space to investigate trends.

(a) Photochem

We make modifications to the default initial atmospheric composition in the Photochem documentation¹⁰ and default atmospheric settings from [Wogan et al. \(2025a\)](#), as this represents

¹⁰"Modern Earth Tutorial": https://github.com/Nicholaswogan/UofA_Photochem_Workshop/tree/main/3_ModernEarthPhotochemistry

the current recommended settings for Earth simulations. Similar to [Cooke et al. \(2022\)](#) and [Cooke et al. \(2023\)](#), we also simultaneously adjust N₂ surface boundary conditions whenever O₂ boundary conditions were modified, ensuring the total atmospheric pressure remained consistent.

Following [Wogan et al. \(2025a\)](#), we adopted a constant relative humidity of 40%, which serves as a tropospheric average in between Earth's modern ranges (between 77% and 20% ([Manabe and Wetherald, 1967](#))). We retain the wet deposition scheme for soluble species (e.g., H₂SO₄, soluble gases), assuming a modern Earth rainfall rate of 1.1×10^{17} molecules cm⁻² s⁻¹ ([Giorgi and Chameides, 1985](#)). The tropopause altitude was set to 1.1×10^6 cm. Upper boundary conditions are uniformly zero escape flux for all species except hydrogen-bearing species (H and H₂), which undergo diffusion limited escape. Lower boundary conditions were chosen to match the globally averaged pre-industrial WACCM6 control case listed in Table 2. For species not explicitly defined in WACCM6, we retain modern Earth surface fluxes and deposition velocities following standard settings described in [Wogan et al. \(2025a\)](#) (see Table 3). Unlike models limited to surface boundary conditions, Photochem allows for fluxes at an altitude, enabling a more realistic emission for volcanic sulfur dioxide (SO₂). To leverage this, we derived a globally averaged vertical emission profile from CESM2 CMIP6 emissions, calculate the flux-weighted centroid height of this profile, and derive a single effective injection height by accounting surface emissions. Several species with lower boundary conditions in WACCM6 runs were not available in Photochem, and were consequently omitted. These include CCl₄, CF₂ClBr, CFC-11eq, CH₂Br₂, CH₃Br, CH₃Cl, and CHBr₃. As these are primarily trace halogens or anthropogenic compounds, their exclusion are not expected to significantly impact the pre-industrial baseline.

Table 3. Boundary conditions for minor species using Earth default settings from [Wogan et al. \(2025a\)](#) where specific WACCM6 boundary conditions were unavailable. For species with emission fluxes specified in WACCM6 (from CESM2 emissions), those are preferred, with deposition velocities are added from Photochem defaults where available.

Species	Flux (molecules cm ⁻² s ⁻¹)	v_{dep} (cm s ⁻¹)
CO	4.98×10^{10}	0.03
NO	2.37×10^9	0.01
NH ₃	1.5×10^{10}	1.0
NO ₂	0	0.05
NO ₃	0	0.05
SO ₂ *	1.43×10^9	1.0
H ₂ S	2.0×10^8	0.015
H ₂ SO ₄	7.0×10^8	1.0
HCN	1.7×10^8	0.13
CH ₃ CN	1.3×10^8	0.13
HNO ₃	0	4.0

* SO₂ flux is at an effective altitude of 1.9 km.

We also ran simulations with varying amounts of atmospheric methane, including 0.1, 0.5, 1, 5 and 10x modern day CH₄. Again, we adjust the partial pressure of N₂ to keep the atmospheric pressure constant. In addition to varying atmospheric composition, we also explore model sensitivities to surface albedo and solar zenith angle by varying them in both a high O₂ (PI) and low O₂ (0.1x PAL) case. Following [Wogan et al. \(2025a\)](#), we set the default surface albedo to 0.2, and vary it between 0.06 and 0.3 to explore the effect of surface albedo on atmospheric composition. Similarly, we adopt the default solar zenith angle of 60 degrees from [Wogan et al. \(2025a\)](#), and vary it between 48 and 60 degrees, which is the range most commonly used in 1D models ([Hu et al. \(2012\)](#); [Moses et al. \(2011\)](#), also see Appendix C in [Tsai et al. \(2021\)](#)).

(i) Iodine Chemistry

We can add iodine stuff from Liu et al. (2025) too by adding I, I₂, OIO (IO₂) etc, and reactions listed in the supplementary info in Liu et al. (2025). Adding molecules and reactions is fairly straightforward, but some iodine oxides don't have publicly available thermodynamic data in Shomate/NASA9 formats. We could potentially make these ourselves from NIST-JANAF tables (e.g Chase (1996) for Iodine oxides)), or talk to a chemist about this. We could also turn off calculations of reverse rates in photochem in *photochem_input_read.f90* globally ¹¹, which bypasses the need for thermodynamic data. This may ruin other calculations, so potentially something we something to talk to Nick Wogan about

(b) Atmos

In Atmos, we again set lower boundary conditions to match the globally averaged pre-industrial WACCM6 control case in table 2, with the exception of species unavailable in the model (CCl₄, CF₂ClBr, CFC-11eq, CH₂Br₂, CH₃Br, CH₃Cl, and CHBr₃). We varied oxygenation states and generate self consistent P-T profiles by coupling the photochemistry and climate codes (*photo* and *clima* respectively). In the *photo* model, we retain default settings from the "Modern Earth Complex" photochemical template ¹², and we retain default settings to use the higher resolution wavelength grid described in Appendix B of Lincowski et al. (2018). Following Lincowski et al. (2018), we also retain the default zenith angle of 60 degrees in *photo* and *clima*, and explore the effect of perturbations from this baseline.

In *clima*, we used the "Modern Earth Complex" climate template ¹³. We set the default surface albedo to 0.2 and turn on the methane setting to ensure radiative accuracy. Crucially, our treatment of water vapour in clime improves upon the constant altitude-independent relative humidity of 40% approximation used in Wogan et al. (2025a). Wogan et al. (2025a) explicitly note that this value represents a tropospheric average of the Manabe and Wetherald (1967) profile, which varies from 77% near the surface to 20% at the tropopause. We approximate this by utilizing the built-in relative humidity parametrization and initialize the surface relative humidity at x% (RH_{surf} = x). This approach aligns methodologically with Lincowski et al. (2018), who also employ the Manabe profile to capture the tropospheric cold trap. However, we draw a distinction in the surface boundary condition. While Lincowski et al. (2018) initialized the profile with a fixed surface relative humidity of 70%, we adopted 77%, to ensure consistency with our Photochem simulations based off of the Wogan et al. (2025a) setup. . When running coupled *photo-clima* runs, we iterate the models back and forth until the change in the ozone column between successive runs converged to 0.5 DU or less.

¹¹ dat%reverse block in https://github.com/Nicholaswogan/photochem/blob/main/src/input/photochem_input_read.f90

¹² Modern Earth Complex Photochemical Template : <https://github.com/VirtualPlanetaryLaboratory/atmos/tree/master/PHOTOCHEM/INPUTFILES/TEMPLATES/ModernEarthComplex>

¹³ Modern Earth Complex climate template : https://github.com/VirtualPlanetaryLaboratory/atmos/blob/master/CLIMA/IO/TEMPLATES/ModernEarthComplex/input_clima.dat

4. Results

(a) Photochem

Figure 4 shows the ozone column value for various oxygenation states. For each oxygen level, the column was calculated using the appropriate PT profile at that oxygenation state. Generally, the column depth of ozone increases with atmospheric oxygenation, except in the 150%PAL atmosphere, where it decreases. This is because when O_2 is too high, the atmosphere "self-shields". At these concentrations, the peak altitude of ozone photolysis shifts upwards, effectively attenuating the photon flux reaching the lower stratosphere, and starving the lower (and denser) atmospheric layers of the atomic oxygen precursors required for efficient ozone production.

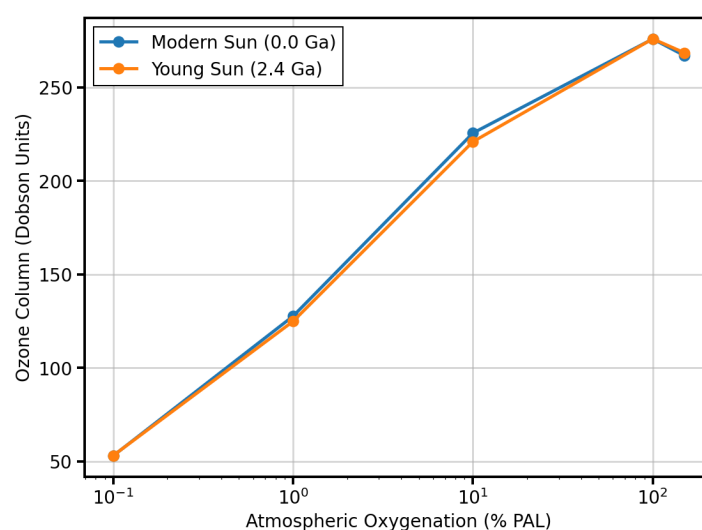


Figure 4. O_3 column densities for various atmospheric oxygen concentrations. For each oxygenation state, the corresponding PT profile was used to compute the oxygen column.

From figure 4, we also note that the effect due to the differing spectra of the young sun only leads to minor differences ozone column depths, with variations being highest when oxygen is at an intermediate (1-10% PAL) level. At low oxygen levels (0.1 % PAL), the abundance of ozone is reactant limited by the scarcity of O_2 . As a result, there is no significant difference in ozone column depths between the spectra of the modern Sun and the young Sun. The greatest sensitivity of Ozone columns to stellar age is observed at 10% PAL, where the amount of ozone is sensitive to the differences in UV fluxes between the faint Sun and the modern Sun. As oxygen approaches and exceeds modern day levels (100-150% PAL), only minor variations exist between the young and modern Suns, suggesting again that the ozone layer is self-shielding and the column depths are limited by saturation rates of the Chapman cycle rather than incident fluxes.

Figure 5 illustrates the impact of an ageing Sun on the ozone profile in more detail. Purely photochemical changes to the ozone layer are small in both the high- O_2 (100% PAL) and low- O_2 (0.1% PAL) cases. The dependence of the total ozone column for a given stellar age is also presented in Figure 5.

(i) Effect of Changing Surface Albedo

As seen in figure 6, increasing surface albedo leads to a decreasing O_3 column.

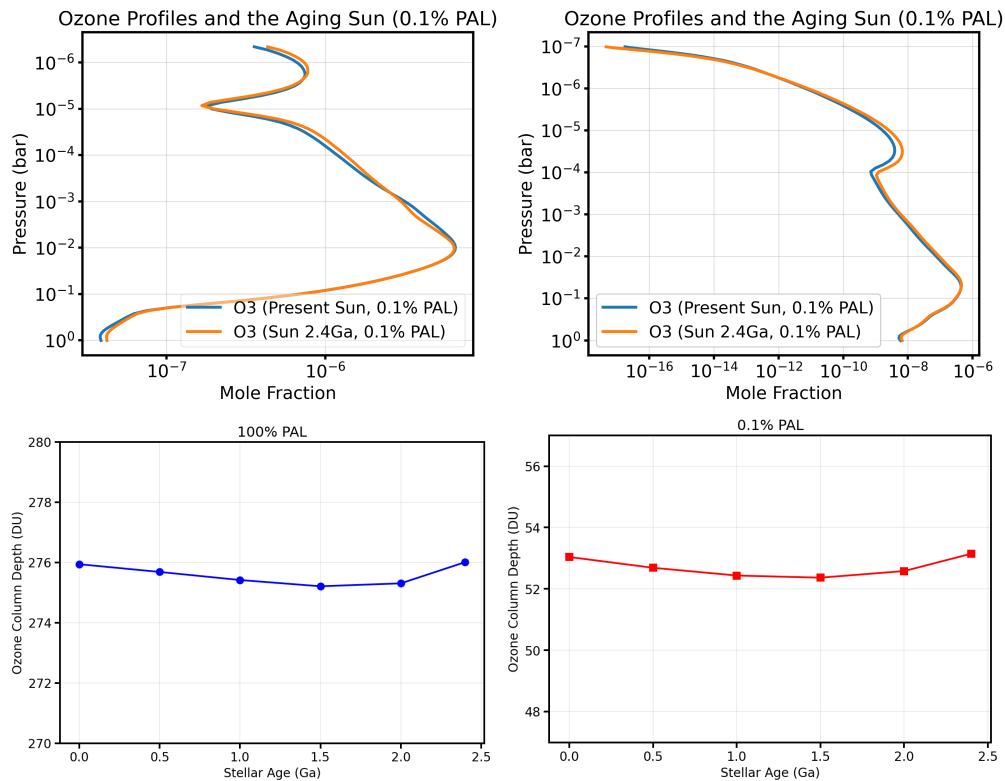


Figure 5. Ozone profiles (top) and the development of the Ozone column (bottom) under an ageing Sun for high O_2 (100% PAL, left) and low O_2 (0.1% PAL, right) atmospheres. For each oxygenation state, the corresponding WACCM6 PT profile was used. Variations due to changes in the spectrum are minimal in both cases.

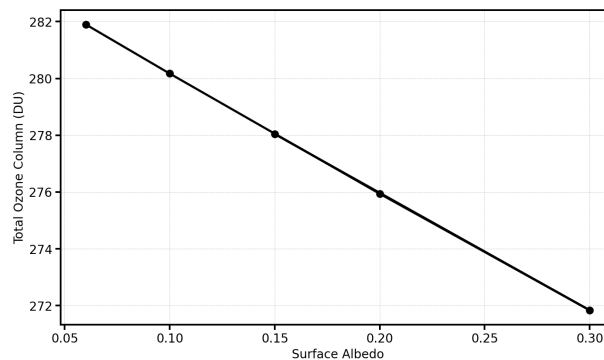


Figure 6. The dependency of O3 column on albedo is shown for 100% PAL case. A few more zenith angle options, 45, 50, 55, 60

(ii) Effect of Changing Zenith Angle

The effect of changing the Zenith Angle is shown in figure 7. Clearly, changes in zenith angle can affect the ozone column significantly. **models must be artificially tuning this to get the right result**

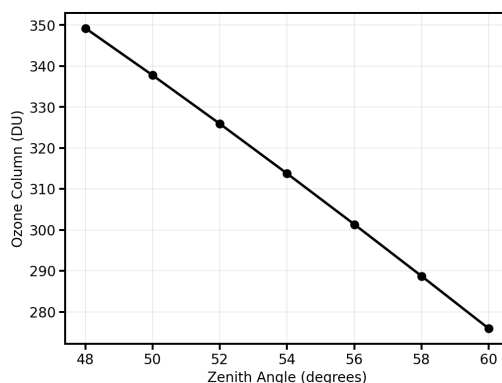


Figure 7. The dependency of O3 column on zenith angle is shown.

5. Methane

The effect of changing the Methane concentration is shown in figure 8.

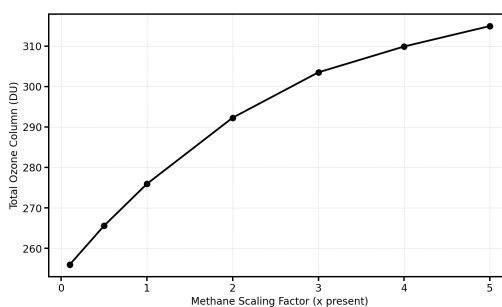


Figure 8. The dependency of O3 column on Methane is shown for 100PAL

6. Discussion

(a) Implications for early Earth

- UV radiation hitting the surface
- Mass extinction events

(b) Exoplanets

- Why it matters for predictions
- Future mission planning
- 2.4 billion years of Earth's history

(c) Future work

- 1D models need to be improved/modified?
- We need to determine the applicability of 1D models. If they are not getting early Earth right, how can we trust models of exoplanets?

- 3D models and model comparison

7. Notes and things to think about

To do list

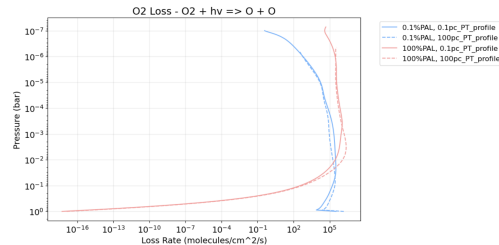


Figure 9.

- Do all of the 1D models implement a surface albedo? If not, how is this affecting the results?
- check how treatments of H₂O change ozone via manabe wetherald parametrization vs constant humidity.
- turning off methane in clima reduces column by around 1DU. eg. 100pc PAL with methane = 204.22 DU without methane = 203.67 DU
- sources in the settings yaml file are misaligned. needs fixing for github
- secondary/tertiary ozone layers
- Model 'tuning'.
- Gaussian quadrature in different models.
- Convergence criteria.
- 10x N₂O sims
- Implications for spectra, can you get similar features for widely different parameters
- Iodine
- Updated Parametrization
- change unit to molecules/cm³ where necessary. integrated loss is in molecules/cm²
- what effect does the ageing sun have on pt profile? i.e couple photochem and clima
- see literature for ozone albedo
- check literature for methane dependence. kasting and donahue evolution of ozone. is the methane increasing at ground level or in the stratosphere?
- check n₂o
- note - running atmos needed changing in fortran code, specifically atmos/photochem/subroutines/output.f line 919, 918. missing comma after 9x. 9x vs 9X seems to make no difference in output. also edited fortran to output o₃ column directly

Kasting Model requires intel compiler

8. Conclusion

- In this work we did... (provide a summary)
- Result 1
- Result 2
- Result 3
- Final sentence/paragraph with outlook

Acknowledgment

Insert the Acknowledgment text here.

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